1.5.1 ▶ ▶ ▶ ▶ Gravitational Potential Energy

▶ The following formula for the gravitational potential of a system of two bodies is required for AP Physics 1, but the derivation is not required.

$$U = -\frac{Gm_1m_2}{r_{12}} \tag{1.69}$$

▶ ▶ ▶ To derive the above formula, we start with the fact that the gravitational force is a conservative force i.e. it can be written as the derivative of some scalar potential[9]

$$F = -\frac{dU}{dr} \tag{1.70}$$

The negative sign might not make sense at first but it is actually added as convention to give the equation more intuitive sense. The negative sign makes it so that conservative forces act in an attempt to decrease potential energy, which is what we experience in everyday life (if you walk up a hill, gravity tries to pull you down). Plugging Newton's Law of Gravitation into eq. 1.69 gives

$$-\frac{dU}{dr} = -\frac{Gm_1m_2}{r^2}$$

$$dU = \frac{Gm_1m_2}{r^2}dr$$

Now, to integrate, we must first define a point to zero potential, since the above expression gives no real indication of where zero is (the reason for +C in indefinite integrals). By convention, we set a point at infinity to zero, since it is common to all systems and is so far removed from all points in the system such that it is (usually) unaffected. However, one could theoretically choose any point they wish. Thus, integrating from infinity to the distance between the two bodies R gives

$$\int_0^{U(R)} dU = \int_\infty^R \frac{Gm_1m_2}{r^2}$$

$$U(R) = \left[-\frac{Gm_1m_2}{r}\right]_\infty^R$$

$$= -\frac{Gm_1m_2}{R} \tag{1.71}$$

which is the general form for gravitational potential energy. Now consider a mass m being raised a height h off the surface of a planet with radius R_E and mass M such that $h << R_E$. Using the above equation, its change in gravitational potential energy is

$$\Delta U = -\frac{GMm}{R_E} - \left(-\frac{GMm}{R_E + h}\right)$$

$$= GMm\left(\frac{1}{R_E + h} - \frac{1}{R_E}\right)$$

$$= GMm\left(\frac{R_E - (R_E + h)}{R_E(R_E + h)}\right)$$

$$= GMm\left(\frac{h/R_E^2}{1 + h/R_E}\right).$$

[9] See 1.3.1

Since $h << R_E$, the expression simply reduces to

$$\begin{aligned} \Delta U &= \frac{GMm}{R_E^2} h \\ &= mgh \end{aligned} \tag{1.72}$$

which is the approximation used for small displacements near the surface of a planet ★